University of North Carolina at Charlotte
Department of Electrical and Computer Engineering
ECGR 3120 Inkjet Printer Simulation

Group Project Report

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Q1

Simulate the motion of the droplet when no voltage is applied across the capacitor. How much time would it take for the droplet to reach the center of the paper?

The droplet is travelling along the x-axis towards the paper. Because there is no acceleration in the x-direction:

$V_X = 20m/s$ $(constant)$ which is independent of V_Y.

The path taken by each droplet regardless of the voltage applied is the same in the x direction. This displacement can be calculated by integrating V_X which gives us:

$S_X = 20 * t\ m$. Solving for time gives us: $\frac{S_X}{20} = t\ sec$. It is given that the distance from droplet gun to the paper is 3mm.

Therefore, the time taken by a droplet to reach the center of the paper with no voltage applied across the capacitor is **0.15 milliseconds**.

Q2

Simulate the act of drawing a vertical line (similar to the letter "I") along the center of the paper at 300 dpi resolution. How much time would it take to draw the letter "I"?